

CLOUD COMPUTING & AWS GLOBAL INFRASTRUCTURE

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Version: 1.1

Date: July 2026

1. Introduction to Cloud Computing

Cloud computing is the on-demand delivery of computing power, database storage, applications, and other IT resources through a cloud services platform via the internet with pay-as-you-go pricing. Instead of buying, owning, and maintaining physical data centers and servers, organizations can access technology services on an as-needed basis from providers like Amazon Web Services (AWS).

2. The Role of NIST

The **National Institute of Standards and Technology (NIST)** is a non-regulatory agency of the United States Department of Commerce. NIST's role in cloud computing was monumental: it provided the definitive, vendor-neutral definition of cloud computing in its Special Publication 800-145. This baseline framework established a standard terminology and criteria that allowed organizations worldwide to evaluate and adopt cloud technologies uniformly.

VISUAL LAYOUT: THE NIST CLOUD FRAMEWORK UMBRELLA

NIST Cloud Computing Definition (SP 800-145)

On-Demand Self-Service

Broad Network Access

Resource Pooling

Rapid Elasticity

Measured Service

3. Essential Characteristics of Cloud Computing (NIST)

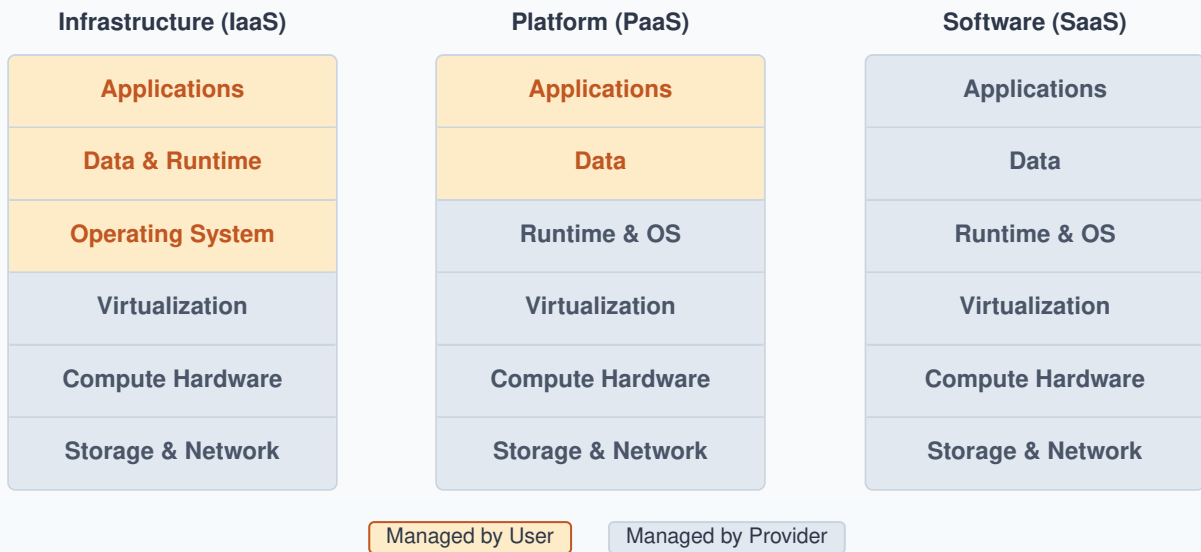
According to the NIST definition, a service must possess all five of the following essential characteristics to be considered a true cloud service:

Characteristic	Detailed Definition	Real-World Example
On-Demand Self-Service	A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.	Launching an EC2 instance via the AWS Management Console in seconds.
Broad Network Access	Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, workstations).	Accessing an application hosted in the cloud from any internet-connected device worldwide.
Resource Pooling	The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.	Multiple independent customer accounts running segregated virtual machines on the same physical host hardware.
Rapid Elasticity	Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear unlimited.	An Auto Scaling group adding instances during a high-traffic flash sale and terminating them when demand subsides.
Measured Service	Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts).	Paying exactly for the number of gigabytes stored per hour in an Amazon S3 bucket.

4. Cloud Service Models

Cloud service models are categorized based on the level of control, flexibility, and management responsibility given to the consumer versus the cloud provider.

VISUAL COMPARISON: SHARED RESPONSIBILITY STACK



Model	Description	Consumer Responsibilities	Examples
IaaS (Infrastructure as a Service)	Offers raw computing primitives like virtual machines, networking infrastructure, and storage. Provides maximum architecture flexibility.	Operating system configuration, patching, applications, network security configuration.	Amazon EC2, Amazon EBS
PaaS (Platform as a Service)	Abstracts the underlying hardware and OS. Allows developers to deploy applications without managing servers or runtimes.	Application code, data, and deployment configurations.	AWS Elastic Beanstalk, AWS Lambda
SaaS (Software as a Service)	A fully functional, end-user business application delivered completely over the web and managed by the vendor.	Basic user configuration, account management, settings.	Amazon WorkDocs, Microsoft 365, Gmail

5. Cloud Deployment Models

Deployment models describe the specific environment types where cloud infrastructure is located and who owns/controls it.

- **Public Cloud:** The infrastructure is owned by a third-party service provider and shared among multiple tenants over the public internet. (e.g., standard AWS offerings).

- ## 6. AWS: Role, Intro, and Brief History

7. AWS Global Infrastructure

- **Availability Zone (AZ):** One or more discrete data centers within an AWS Region. Each AZ features redundant power, networking, and cooling. AZs are separated by several miles to mitigate localized disasters (floods, fires), yet connected via high-speed, private fiber optic links to guarantee low-latency connectivity.
- **Data Centers:** The physical facilities containing the actual servers, storage arrays, switches, and core infrastructure. An individual AZ is comprised of one or more of these data centers.

8. Security in AWS Data Centers

AWS handles physical and environmental infrastructure security under the Shared Responsibility Model, employing strict defense-in-depth principles:

- **Physical Access Control:** Data centers are non-descript buildings with strictly gated and monitored perimeters. Access requires multi-factor authentication, biometric scanning, and approved security logging.
- **Environmental Protections:** Fully redundant electrical infrastructure (UPS systems and multi-day diesel backup generators) paired with custom atmospheric cooling units ensure uninterrupted operational capabilities.
- **Decommissioning and Destruction:** Legacy storage media devices containing customer data undergo rigorous multi-step destruction procedures, including degaussing (demagnetization) and physical shredding before final disposal.

9. Factors to Choose an AWS Region

Architects must evaluate four primary operational vectors when selecting an optimal deployment Region:

Factor	Architectural Context and Considerations
Compliance & Data Sovereignty	Many regional legal mandates require explicit data residency restrictions (e.g., GDPR in Europe). Applications must be deployed inside regions matching those legal geographic boundaries.
Latency & Proximity	Deploying applications in close geographic proximity to the primary user demographic decreases network packet round-trip times and increases overall app responsiveness.
Service Availability	Not all specialized or newly released AWS features are rolled out to every single global region simultaneously. Core services exist everywhere, but advanced features require explicit validation.
Cost and Pricing	AWS rates differ slightly per region based on variable regional overheads like power costs, taxes, land acquisition, and local fiber connectivity agreements.